

Dockerized Java Web Application Deployment Using Maven WAR and Apache Tomcat

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Command Prompt

C:\Users\sreev>mvn archetype:generate -DgroupId=com.demo -DartifactId=tomcatdemo -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 17 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 24 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1 kB at 28 kB/s)
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 29 MB/s)
[INFO]
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\sreev
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: tomcatdemo
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\sreev\tomcatdemo
[INFO]
[INFO] BUILD SUCCESS
[INFO] -----
```

```
Command Prompt

Directory of C:\Users\sreev\tomcatdemo

09-05-2026 17:20 <DIR>      .
09-05-2026 17:20 <DIR>      ..
09-05-2026 17:20          734 pom.xml
09-05-2026 17:20 <DIR>      src
                1 File(s)      734 bytes
                3 Dir(s)  38,773,989,376 bytes free

C:\Users\sreev\tomcatdemo>tree /f
Folder PATH listing for volume OS
Volume serial number is 00000049 0CE8:3131
C:.
|_ pom.xml
|_ src
|   |_ main
|       |_ resources
|           |_ webapp
|               |_ index.jsp
|               |_ WEB-INF
|                   |_ web.xml

C:\Users\sreev\tomcatdemo>notepad src\main\webapp\index.jsp

C:\Users\sreev\tomcatdemo>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:tomcatdemo >-----
[INFO] Building tomcatdemo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ tomcatdemo ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ tomcatdemo ---
```

```
C:\Users\sreev\tomcatdemo>type dockerfile
FROM tomcat:10.1
COPY target/tomcatdemo.war /usr/local/tomcat/webapps/
EXPOSE 8080
CMD ["catalina.sh","run"]
C:\Users\sreev\tomcatdemo>
```

```
Command Prompt
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:tomcatdemo >-----
[INFO] Building tomcatdemo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ tomcatdemo ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ tomcatdemo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ tomcatdemo ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ tomcatdemo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sreev\tomcatdemo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ tomcatdemo ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ tomcatdemo ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ tomcatdemo ---
[INFO] Packaging webapp
[INFO] Assembling webapp [tomcatdemo] in [C:\Users\sreev\tomcatdemo\target\tomcatdemo]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\sreev\tomcatdemo\src\main\webapp]
[INFO] Building war: C:\Users\sreev\tomcatdemo\target\tomcatdemo.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.266 s
[INFO] Finished at: 2026-05-09T17:27:43+05:30
[INFO] -----
```

```
Command Prompt
C:\Users\sreev\tomcatdemo>docker build -t tomcatdemo:v1 .
[+] Building 115.2s (8/8) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile                0.1s
=> => transferring dockerfile: 148B                                0.0s
=> [internal] load metadata for docker.io/library/tomcat:10.1     4.2s
=> [auth] library/tomcat:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                   0.2s
=> => transferring context: 1.80kB                                    0.1s
=> [1/2] FROM docker.io/library/tomcat:10.1@sha256:04cbc0cc725471cddc68ca8c85b61106c05b2361f93a0f9566c1649d062 109.5s
=> => resolve docker.io/library/tomcat:10.1@sha256:04cbc0cc725471cddc68ca8c85b61106c05b2361f93a0f9566c1649d06295 0.1s
=> => sha256:f1bf71542dc83f87b76ebb92c0f29ad9a7bbce786895ac6217795f5b7b62878b 14.53MB / 14.53MB 6.1s
=> => sha256:e7a1bbef824a22bbdbcd74d9b9806c02cf2bdab21d0ff53e2ff0083820ef2d2 139B / 139B 1.3s
=> => sha256:d92a694d8aa025f026ea647e2324c8daccda22c915159e8672c4a5f3e6861c91 2.28kB / 2.28kB 1.7s
=> => sha256:b38893372039c6251d8c3f63d95d127b0ce85aa0cd7250e1c5da23898f264e01 92.71MB / 92.71MB 107.1s
=> => sha256:97c47a5c1a2f55ee2696423e5c0582023889a363dd5a3ec51a2633d6d340d973 17.46MB / 17.46MB 21.3s
=> => extracting sha256:97c47a5c1a2f55ee2696423e5c0582023889a363dd5a3ec51a2633d6d340d973 0.6s
=> => extracting sha256:b38893372039c6251d8c3f63d95d127b0ce85aa0cd7250e1c5da23898f264e01 1.4s
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 0.0s
=> => extracting sha256:492a694d8aa025f026ea647e2324c8daccda22c915159e8672c4a5f3e6861c91 0.0s
=> => extracting sha256:e7a1bbef824a22bbdbcd74d9b9806c02cf2bdab21d0ff53e2ff0083820ef2d2 0.0s
=> => extracting sha256:f1bf71542dc83f87b76ebb92c0f29ad9a7bbce786895ac6217795f5b7b62878b 0.3s
=> [2/2] COPY target/tomcatdemo.war /usr/local/tomcat/webapps/ 0.5s
=> => exporting to image 0.5s
=> => exporting layers 0.2s
=> => exporting manifest sha256:7a06946c97a52b62a24baa68409d839b850becfe40d80e6d8a1492242e7c08b3 0.0s
=> => exporting config sha256:d3a28618a7293de133a281a92a1358cfccf82a50861d31a06f1568c3d9080d3 0.0s
=> => exporting attestation manifest sha256:40c64ecfcd6f0b38d9a656b72cce12fa44f837fc9ffefef2c0a9d72c6b6929d1 0.1s
=> => exporting manifest list sha256:7aefa56a847a554f8efb0847a39cb95826c080a84e574f12c53d72e3ec0a1376 0.0s
=> => naming to docker.io/library/tomcatdemo:v1 0.0s
=> => unpacking to docker.io/library/tomcatdemo:v1 0.1s

C:\Users\sreev\tomcatdemo>docker run -d --name tomcatapp -p 8085:8080 tomcatdemo:v1
82f039e944253c4fdddf445140f3927ccff96b84e6c8c72c899ef5f5cbc53dfc

C:\Users\sreev\tomcatdemo>
```



Hello from Tomcat Docker

WAR deployment successful